

TASK 3 — SQL QUERY BOOK

E-commerce Sales Analytics • Executed SQL + Output Evidence

01. Basic SELECT + filtering

```
SELECT order_id, order_date, product_category, region, revenue
FROM sales
WHERE revenue > 1500
ORDER BY revenue DESC
LIMIT 10;
```

Output:

order_id	order_date	product_category	region	revenue
10810	2024-03-20	Electronics	South	4,119.33
14884	2035-05-16	Electronics	South	4,083.17
12215	2028-01-24	Electronics	South	4,055.29
13102	2030-06-29	Beauty	East	4,005.93
13687	2032-02-04	Clothing	North	3,928.63
10158	2022-06-07	Electronics	North	3,917.54
13350	2031-03-04	Electronics	West	3,898.48
10322	2022-11-18	Electronics	West	3,893.11
10662	2023-10-24	Clothing	East	3,868.11
14306	2033-10-15	Beauty	South	3,807.92

02. Regional revenue

```
SELECT region,  
COUNT(*) AS orders,  
SUM(quantity) AS units_sold,  
ROUND(SUM(revenue),2) AS total_revenue,  
ROUND(AVG(revenue),2) AS avg_order_revenue  
FROM sales  
GROUP BY region  
ORDER BY total_revenue DESC;
```

Output:

region	orders	units_sold	total_revenue	avg_order_revenue
West	1289	5260	1,345,582.16	1,043.90
North	1276	5164	1,281,508.45	1,004.32
South	1208	4884	1,246,640.90	1,031.99
East	1227	4916	1,236,044.23	1,007.37

03. Category performance

```
SELECT product_category,  
COUNT(*) AS orders,  
SUM(quantity) AS units_sold,  
ROUND(SUM(revenue),2) AS revenue,  
ROUND(AVG(customer_rating),2) AS avg_rating  
FROM sales  
GROUP BY product_category  
ORDER BY revenue DESC;
```

Output:

product_category	orders	units_sold	revenue	avg_rating
Electronics	1777	7109	1,829,899.22	2.95
Clothing	1531	6171	1,531,931.72	3.01
Home	969	3949	982,083.92	2.94
Beauty	723	2995	765,860.88	3.01

04. Payment method mix

```
SELECT payment_method,  
COUNT(*) AS orders,  
ROUND(100.0*COUNT(*)/(SELECT COUNT(*) FROM sales),2) AS order_share_pct,  
ROUND(SUM(revenue),2) AS revenue  
FROM sales  
GROUP BY payment_method  
ORDER BY revenue DESC;
```

Output:

payment_method	orders	order_share_pct	revenue
Card	2270	45.40	2,366,248.20
COD	1774	35.48	1,788,408.44
Wallet	956	19.12	955,119.10

05. Monthly revenue trend

```
SELECT strftime('%Y-%m', order_date) AS month,
COUNT(*) AS orders,
ROUND(SUM(revenue),2) AS revenue
FROM sales
GROUP BY month
ORDER BY month;
```

Output:

month	orders	revenue
2022-01	31	35,624.89
2022-02	28	29,908.89
2022-03	31	30,309.28
2022-04	30	30,781.96
2022-05	31	41,629.24
2022-06	30	29,847.33
2022-07	31	28,686.24
2022-08	31	29,186.45
2022-09	30	28,791.07
2022-10	31	29,265.55
2022-11	30	32,329.49
2022-12	31	25,463.91

06. HAVING: high-value regions

```
SELECT region,
ROUND(SUM(revenue),2) AS revenue,
ROUND(AVG(customer_rating),2) AS avg_rating
FROM sales
GROUP BY region
HAVING SUM(revenue) > (SELECT AVG(region_revenue)
FROM (SELECT SUM(revenue) AS region_revenue
FROM sales GROUP BY region))
ORDER BY revenue DESC;
```

Output:

region	revenue	avg_rating
West	1,345,582.16	2.96
North	1,281,508.45	2.99

07. Average revenue per user

```
SELECT ROUND(AVG(customer_revenue),2) AS avg_revenue_per_customer
FROM (
  SELECT customer_id, SUM(revenue) AS customer_revenue
  FROM sales
  GROUP BY customer_id
) t;
```

Output:

avg_revenue_per_customer

5,166.61

08. Customer leaderboard

```
SELECT customer_id,
COUNT(*) AS orders,
ROUND(SUM(revenue),2) AS lifetime_revenue,
ROUND(AVG(customer_rating),2) AS avg_rating
FROM sales
GROUP BY customer_id
ORDER BY lifetime_revenue DESC
LIMIT 10;
```

Output:

customer_id	orders	lifetime_revenue	avg_rating
1,663.00	14.00	17,680.69	2.75
1,955.00	10.00	16,265.15	3.49
1,675.00	11.00	15,036.46	2.98
1,276.00	11.00	15,023.71	3.04
1,647.00	11.00	14,894.97	3.03
1,804.00	10.00	14,330.98	3.04
1,193.00	7.00	14,151.65	3.27
1,836.00	9.00	14,110.78	3.31
1,957.00	12.00	13,956.95	3.18
1,735.00	8.00	13,146.25	3.29

09. INNER JOIN: sales + customer segment

```
SELECT c.customer_segment,
COUNT(s.order_id) AS orders,
ROUND(SUM(s.revenue),2) AS revenue
FROM sales s
INNER JOIN customers c ON s.customer_id = c.customer_id
GROUP BY c.customer_segment
ORDER BY revenue DESC;
```

Output:

customer_segment	orders	revenue
Value	1696	1,732,764.15
Premium	1659	1,728,167.10
Standard	1645	1,648,844.49

10. LEFT JOIN: category + margin reference

```
SELECT p.product_category,  
p.category_margin,  
COUNT(s.order_id) AS orders,  
ROUND(SUM(s.revenue),2) AS revenue,  
ROUND(SUM(s.revenue)*p.category_margin,2) AS est_gross_profit  
FROM products p  
LEFT JOIN sales s ON p.product_category = s.product_category  
GROUP BY p.product_category, p.category_margin  
ORDER BY est_gross_profit DESC;
```

Output:

product_category	category_margin	orders	revenue	est_gross_profit
Clothing	0.29	1531	1,531,931.72	444,260.20
Electronics	0.22	1777	1,829,899.22	402,577.83
Home	0.27	969	982,083.92	265,162.66
Beauty	0.34	723	765,860.88	260,392.70

11. Subquery: above-average orders

```
SELECT order_id, customer_id, product_category, revenue
FROM sales
WHERE revenue > (SELECT AVG(revenue) FROM sales)
ORDER BY revenue DESC
LIMIT 10;
```

Output:

order_id	customer_id	product_category	revenue
10810	1890	Electronics	4,119.33
14884	1146	Electronics	4,083.17
12215	1113	Electronics	4,055.29
13102	1699	Beauty	4,005.93
13687	1831	Clothing	3,928.63
10158	1779	Electronics	3,917.54
13350	1082	Electronics	3,898.48
10322	1608	Electronics	3,893.11
10662	1550	Clothing	3,868.11
14306	1283	Beauty	3,807.92

12. CTE: delivery experience

```
WITH delivery AS (  
  SELECT region,  
    COUNT(*) AS orders,  
    ROUND(AVG(delivery_days),2) AS avg_delivery_days,  
    ROUND(AVG(customer_rating),2) AS avg_rating  
  FROM sales  
  GROUP BY region  
)  
SELECT * FROM delivery  
ORDER BY avg_delivery_days ASC;
```

Output:

region	orders	avg_delivery_days	avg_rating
West	1289	6.08	2.96
East	1227	6.10	2.97
North	1276	6.12	2.99
South	1208	6.17	2.97

13. Window function: category revenue share

```
WITH cat AS (  
  SELECT product_category, SUM(revenue) AS revenue  
  FROM sales GROUP BY product_category  
)  
SELECT product_category,  
  ROUND(revenue,2) AS revenue,  
  ROUND(100.0*revenue/SUM(revenue) OVER ( ),2) AS revenue_share_pct  
FROM cat  
ORDER BY revenue DESC;
```

Output:

product_category	revenue	revenue_share_pct
Electronics	1,829,899.22	35.81
Clothing	1,531,931.72	29.98
Home	982,083.92	19.22
Beauty	765,860.88	14.99

14. Ranking: top 3 categories by region

```
WITH x AS (  
  SELECT region, product_category,  
         SUM(revenue) AS revenue  
  FROM sales  
  GROUP BY region, product_category  
)  
ranked AS (  
  SELECT *,  
         DENSE_RANK() OVER (PARTITION BY region ORDER BY revenue DESC) AS rnk  
  FROM x  
)  
SELECT region, product_category, ROUND(revenue,2) AS revenue, rnk  
FROM ranked  
WHERE rnk <= 3  
ORDER BY region, rnk;
```

Output:

region	product_category	revenue	rnk
East	Electronics	411,803.35	1
East	Clothing	402,582.60	2
East	Home	246,749.17	3
North	Electronics	472,450.93	1
North	Clothing	364,659.43	2
North	Home	258,184.77	3
South	Electronics	487,055.68	1
South	Clothing	348,363.14	2
South	Home	220,148.10	3
West	Electronics	458,589.26	1
West	Clothing	416,326.55	2
West	Home	257,001.88	3

15. Data quality audit

```
SELECT
SUM(CASE WHEN order_id IS NULL THEN 1 ELSE 0 END) AS null_order_id,
SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END) AS null_customer_id,
SUM(CASE WHEN revenue IS NULL THEN 1 ELSE 0 END) AS null_revenue,
SUM(CASE WHEN revenue < 0 THEN 1 ELSE 0 END) AS negative_revenue,
COUNT(DISTINCT order_id) AS unique_orders
FROM sales;
```

Output:

null_order_id	null_customer_id	null_revenue	negative_revenue	unique_orders
0	0	0	0	5000

Optimization SQL

```
CREATE INDEX idx_sales_customer ON sales(customer_id);  
CREATE INDEX idx_sales_region_date ON sales(region, order_date);  
CREATE INDEX idx_sales_category ON sales(product_category);
```

This query book contains the executable SQL used for the report. The results shown are generated from the supplied 5,000-row dataset.